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AI and the distribution of income between capital and labour

Antonio Minniti, Klaus Prettnner, Francesco Venturini, David Bloom / 3 Mar 2026

The swift rise of artificial intelligence is raising fundamental concerns about the future of work. This column uses data from 238 regions across 21 European countries to examine how AI-related innovation influences the distribution of income between labour and capital, and among different skill classes of labour. Regions with more intense AI patenting tend to experience a decline in the labour share of income, especially in areas with a strong industrial base, indicating that AI acts as a capital-biased innovation. Without appropriate policy intervention, this trend could exacerbate existing inequalities.

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The swift rise of artificial intelligence (AI) is reshaping production systems and raising fundamental concerns about the future of work (e.g. Korinek and Stiglitz 2019, Webb 2019, Prytkova et al. 2024). Will AI mainly support human labour or render it redundant? Who will capture the benefits of this transformation – workers or capital owners? And among workers, who will fare relatively better – the more- or less-skilled?

Although these questions dominate much of the public debate on AI, robust empirical evidence remains limited. In a recent article (Minniti et al. 2025) we offer new insights into how AI-related innovation influences the distribution of income between labour and capital, and among different skill classes of labour, in European regions.

The results show that regions with more intense AI patenting tend to experience a decline in the labour share of income, especially in areas with a strong industrial base. This pattern indicates that AI acts as a capital-biased innovation, shifting the returns from technological progress increasingly towards capital.

If not addressed through targeted policies, this trend could exacerbate existing inequalities and pose long-term challenges to social cohesion in advanced economies.

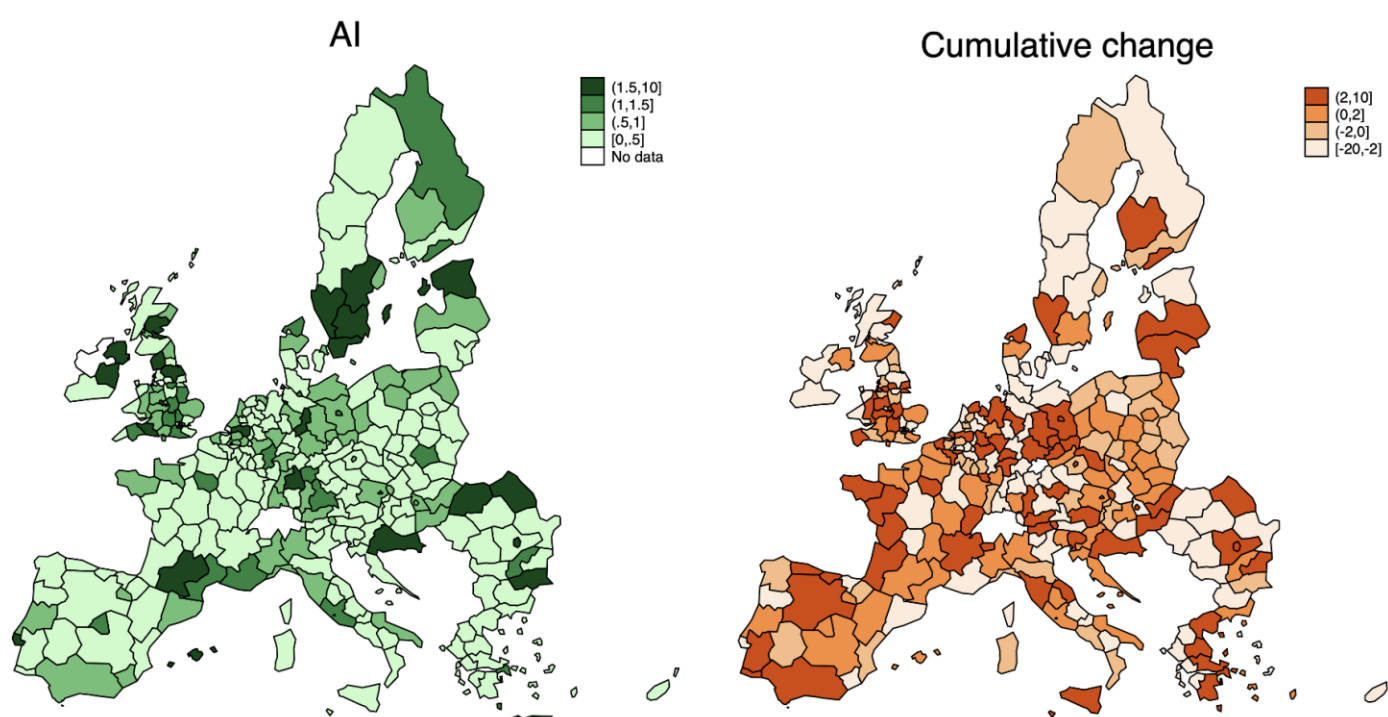
AI innovation, skills, and the changing distribution of income

We examine the relationship between AI-driven innovation and the labour share using data from 238 regions across 21 European countries, spanning the period 2000–2017.¹ This regional lens is essential, as differences in industrial structure, educational attainment, and local innovation dynamics strongly mediate the effects of technological change.

We measure AI innovation through a newly constructed dataset of AI-related patents, including technologies such as machine learning, natural language processing, image-processing methods, and so on. These data are linked to detailed regional indicators on employment, wages, and value added, disaggregated by skill level.

The findings reveal a robust and statistically significant negative association between AI patenting intensity and the labour income share. A doubling in AI patent intensity is associated with a 0.5 to 1.6 percentage point reduction in the labour share. Figure 1 illustrates this development, displaying AI intensity across European regions (left panel) and the cumulative change in the labour share between 2000 and 2017 (right panel). In regions with higher AI intensity, the labour income share tended to decrease during the overall time span.

Figure 1



Note: The left-hand side of the graph reports regional AI patenting intensity, defined as technologically revealed comparative advantage, calculated as the average count of AI patents relative to that of other patents filed in the region between 2000 and 2017. The resulting value is then expressed as a ratio to the same measure for the other regions. If the index is greater than 1, the region is specialized in AI patenting. The right-hand side shows the cumulative change in the labour share between 2000 and 2017, expressed in percentage points.

Crucially, the decline in the labour share is not uniform across the workforce. We find that medium- and high-skilled workers experience the strongest reductions in their income share, primarily driven by wage compression as opposed to changes in employment.² By contrast, the labour share of low-skilled workers declines less – largely because of mild employment growth in this segment, which partly offsets stagnant or falling wages. These patterns suggest that AI is reshaping the labour market not only between labour and capital, but also within labor, by redistributing returns across skill groups (see also Bloom et al. 2024 and 2025 for a theoretical framework that is consistent with these results).

Importantly, we do not observe proportional gains in labour productivity that might offset these distributional shifts. In this context, AI not only raises efficiency but also reallocates the gains from innovation, amplifying the share of income accruing to capital.

From routine substitution to skill compression: Rethinking polarisation

While the decline in the labour share began long before the emergence of AI, the evidence suggests that AI innovation is reinforcing and accelerating this trend. Historically, technological progress has often complemented skilled labour while replacing routine, low-skill tasks. By contrast, modern AI is increasingly able to replicate cognitive work, expanding the scope of automation into high- and medium-skill occupations.

This shift is particularly visible in the service economy. AI applications are now used in law, finance, logistics, public administration, and other white-collar domains previously considered resistant to automation. As a result, the current wave of innovation does not conform to the classical narrative of low-skill job displacement followed by skill upgrading. Instead, we observe signs of wage compression at the top and middle of the skill distribution.

The net effect is a new form of labour market polarisation, one that does not necessarily benefit high-skilled workers as in past waves of technological change. Rather, the evidence points to a redistribution of economic value away from labour across the board, with particularly adverse effects for those previously insulated from technological disruption. This occurs even against the backdrop that AI raises average productivity and economic growth (e.g., Acemoglu 2025).

In the European context, where concerns over wage stagnation, inequality, and political discontent are mounting, this evolving dynamic demands close attention.

Policy implications

The economic effects of AI diffusion are not deterministic – the distributional outcomes will depend on how policymakers respond to the deployment and increasing use of this technology. To mitigate the adverse labour effects, investing in human capital is crucial, particularly through the development of lifelong learning systems and vocational programmes that equip workers with the skills to transition into AI-complementary roles.

In tandem, tax systems must adapt to the shifting balance between labour and capital. As capital increasingly captures the returns from production, fiscal frameworks could be developed to ensure equity and revenue sustainability. Potential policies could reduce labour taxation and move to a (higher) taxation of pollution, which has negative externalities that would be efficient to reduce anyway; a (higher) taxation of land, which is fixed in supply so that its taxation is non-distortionary; and a (higher) taxation of consumption, which will continue to grow in the presence of AI-driven economic growth (for a detailed discussion of such policies see Prettnner and Bloom 2020, especially Chapter 7). Moreover, to avoid reinforcing geographic disparities, economic policy could actively promote the diffusion of AI technologies across regions and sectors. Broadening AI adoption can help ensure that productivity gains are not confined to a few innovation hubs but instead contribute to inclusive and regionally balanced growth.

If governed effectively, AI could appreciably raise aggregate economic wellbeing. Without appropriate policy intervention, however, it may reinforce structural divides – between labour and capital and between core and peripheral regions.

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Footnotes

1. According to the Nomenclature of Territorial Units for Statistics (NUTS) of the EU, the data refer to the NUTS-2 level (basic regions).
2. The skill types are defined using the ISCED classification system, which segments educational attainment into three categories: Low (early childhood education; primary education. lower secondary education), Medium (upper secondary education. post-secondary non-tertiary education), and High (short-cycle tertiary education; bachelor's degree or equivalent tertiary education level).

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LABOUR MARKETS MACROECONOMIC POLICY
PRODUCTIVITY AND INNOVATION

KEYWORDS

ARTIFICIAL INTELLIGENCE LABOUR SHARE OF INCOME

SHARE

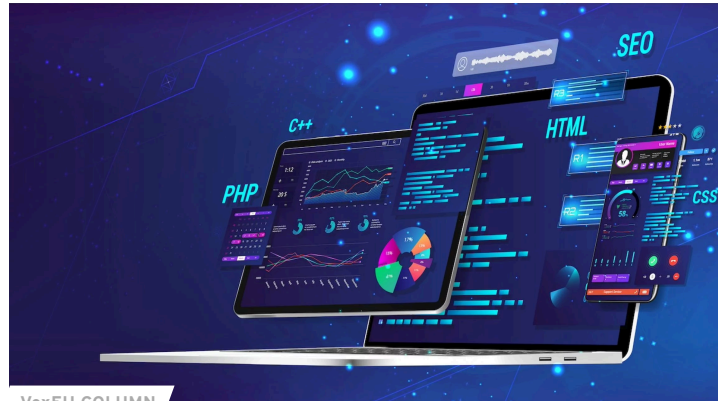


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